

# Detection and Classification of Small-sized UAVs and Birds in Sparse LiDAR Point Cloud

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# Motivation and Problem Statement

- Consumer drones pose safety and privacy risks near critical infrastructure
- Existing RADAR and camera solutions have limitations for 3D detection
- Rosette scanning LiDAR offers spatial data but produces sparse point clouds
- Challenge: Distinguish UAVs from birds and filter false positives
- Goal: Real-time, robust detection and classification using only LiDAR data

## Related Work: LiDAR-based UAV Detection

- Prior methods use octomap-based background subtraction, but struggle with dynamic scenes
- Camera-based and range image methods require dense data or additional sensors
- Movement pattern analysis can misclassify birds as threats
- Our approach: Only LiDAR, up to 100m range, robust to dynamic environments

# Proposed Pipeline Overview

- Semantic segmentation removes background points
- Foreground points clustered via K-Means
- Nearest Neighbour Search (NNS) refines clusters
- Clusters classified as drone or bird using modified PointNet

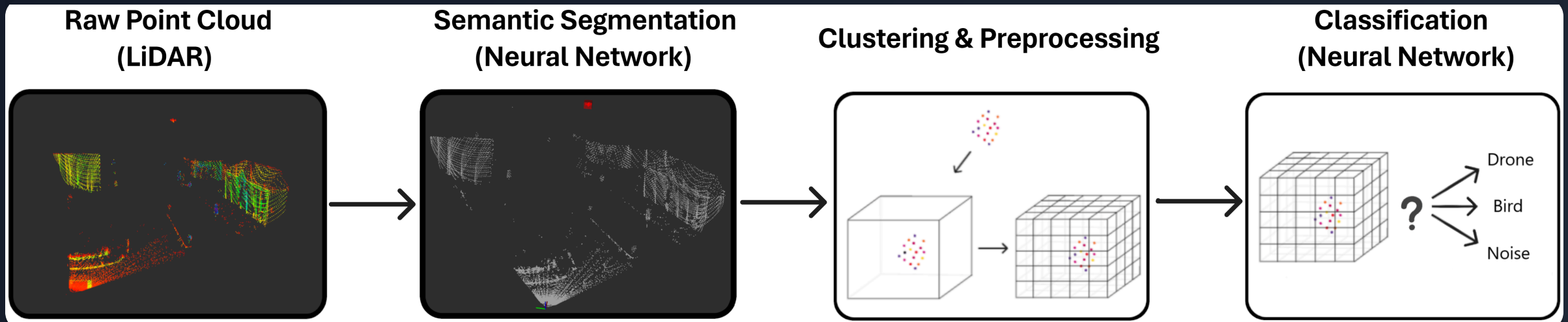


Fig. 1. Outline of the proposed solution: The raw point cloud stream, that is provided by a rosette scanning LiDAR serves as input for an attention-based neural network that removes both the static and dynamic background elements from it. The remaining foreground points then undergo a clustering and a unique preprocessing step in order to be classified by a slightly modified PointNet network.

# Attention-based Semantic Segmentation Network

- Spatial transformer (T-Net) aligns point cloud to a uniform space
- Multi-head attention encoder captures global context
- Points embedded with learnable features, no voxelization
- 1D convolutional layers assign background/foreground labels

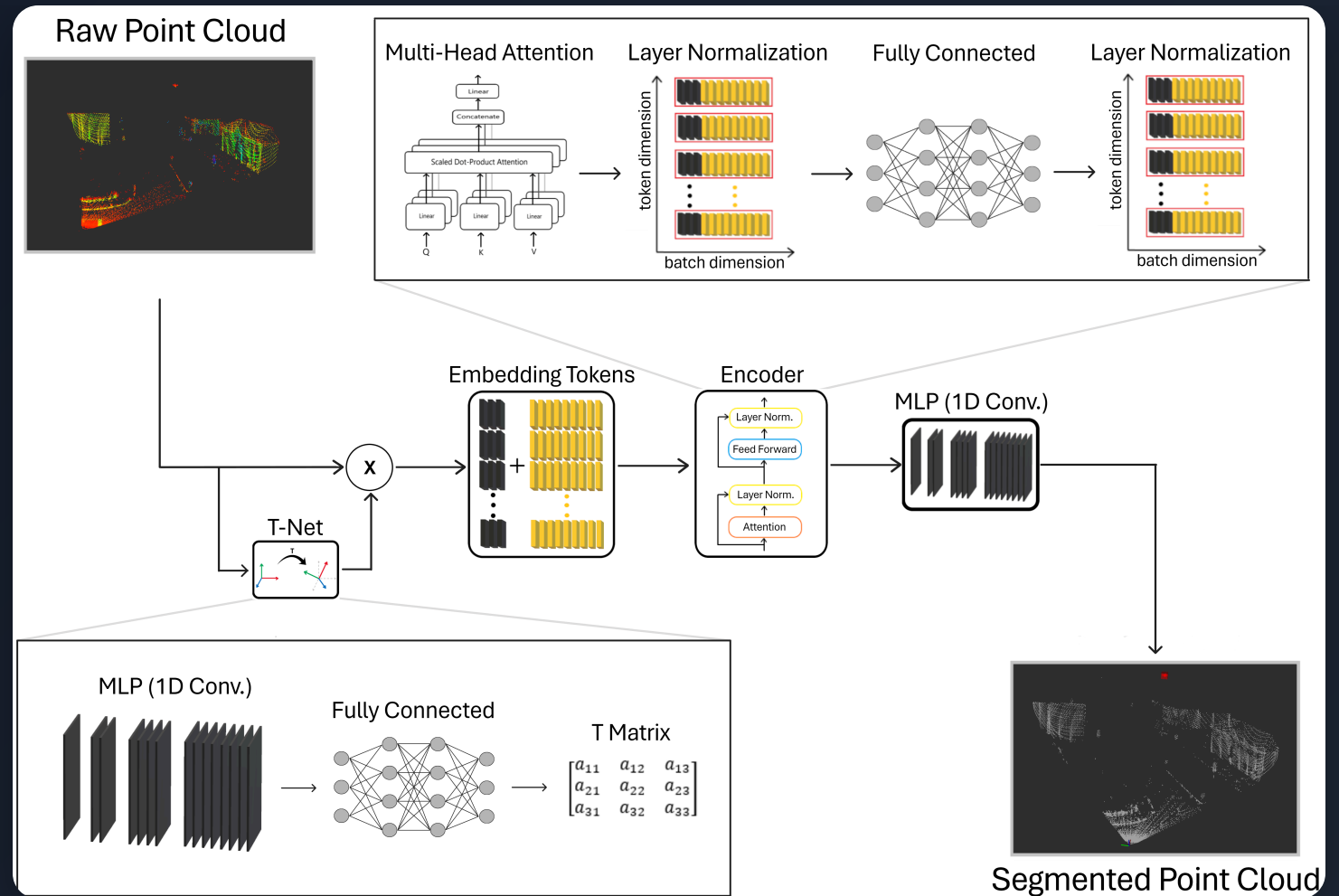


Fig. 2. Overview of the semantic segmentation network: the point cloud is transformed into a transformation invariant space, where each of the points influence is calculated with relevance to all the other points through the Multi-Head attention [17] mechanism and finally, all of them are assigned with a class determined by an MLP.

# Preprocessing and Classification

- Background points removed after segmentation
- Clusters with fewer than 3 points discarded as noise
- NNS recovers missed foreground points
- Clusters voxelized into uniform grids for classification
- Modified PointNet classifies clusters as drone, bird, or unknown

# Real-Time Performance: Component-wise Speed

- System processes up to 60 FPS on 1100 point clouds
- All components operate within real-time constraints
- Preprocessing is the most time-consuming step

| Component      | Mean (ms) | Worst-case (ms) |
|----------------|-----------|-----------------|
| Segmentation   | 4.29      | 6.96            |
| Clustering     | 1.65      | 8.86            |
| Preprocessing  | 4.11      | 16.82           |
| Classification | 6.44      | 8.14            |

# Static vs Dynamic Background Removal Speed

- Dynamic segmentation (ours) is significantly faster
- Lower runtime deviation compared to static method
- Enables robust operation in dynamic environments

| Method         | Mean (ms) | Worst-case (ms) |
|----------------|-----------|-----------------|
| Dynamic (ours) | 4.29      | 6.96            |
| Static [5]     | 16.49     | 57.27           |

# Semantic Segmentation Performance

- Balanced F1-score used due to class imbalance
- F1-score of 0.78 achieved on test dataset
- Low false positive rate enables reliable object recovery

| Metric    | Value  |
|-----------|--------|
| Precision | 0.7634 |
| Recall    | 0.7949 |
| F-Measure | 0.7787 |

# Classification Network Results

- Classification accuracy: 81.5% on balanced drone/bird dataset
- Confusion matrix shows reliable discrimination
- Performance evaluated in end-to-end pipeline context

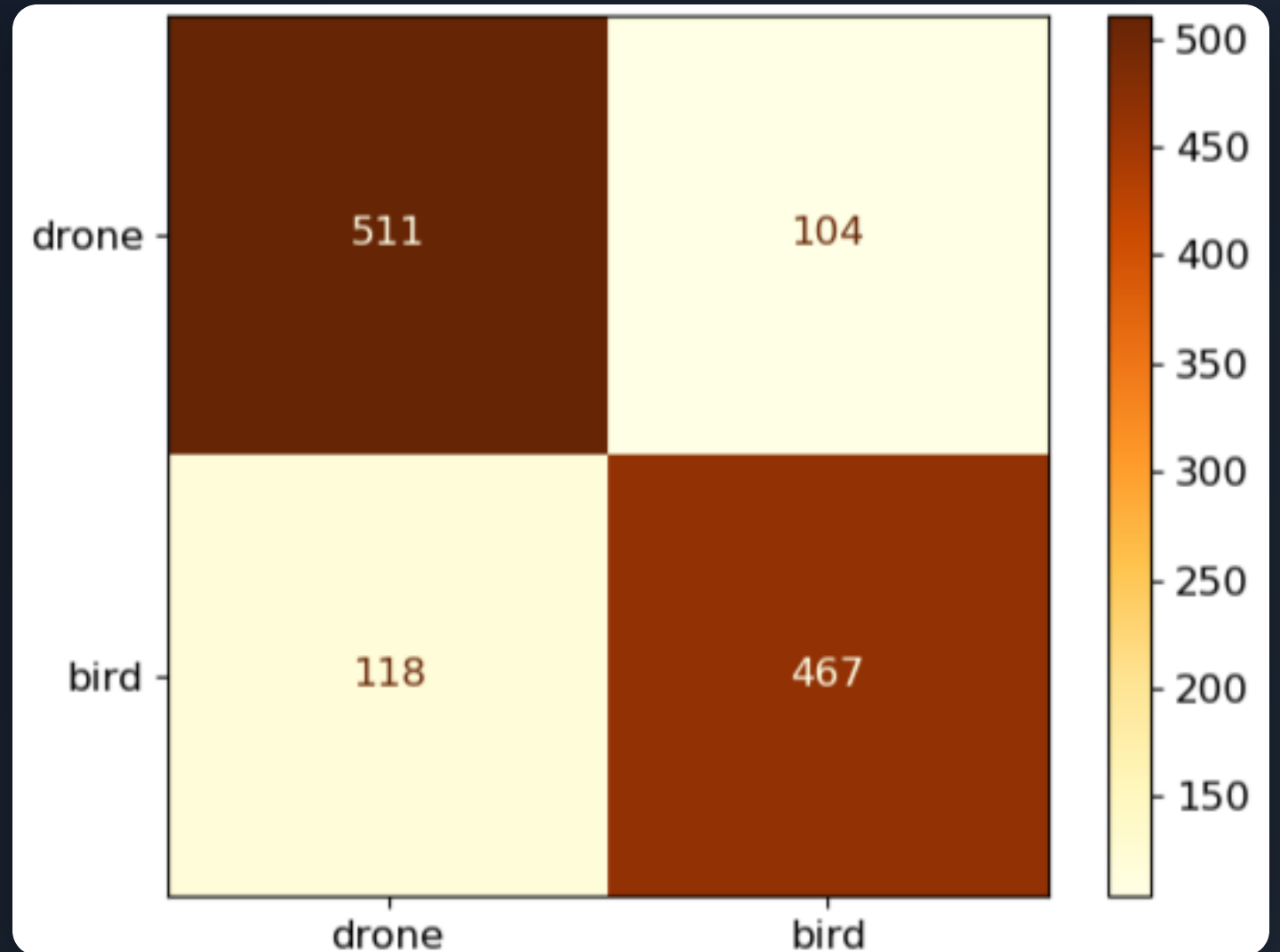


Fig. 3. Confusion matrix on the test dataset for the classification network.

# Distance vs Classification Accuracy

- Classification accuracy remains stable across distances
- Sparse training samples aid generalization for longer ranges
- Effective range up to 100 meters demonstrated

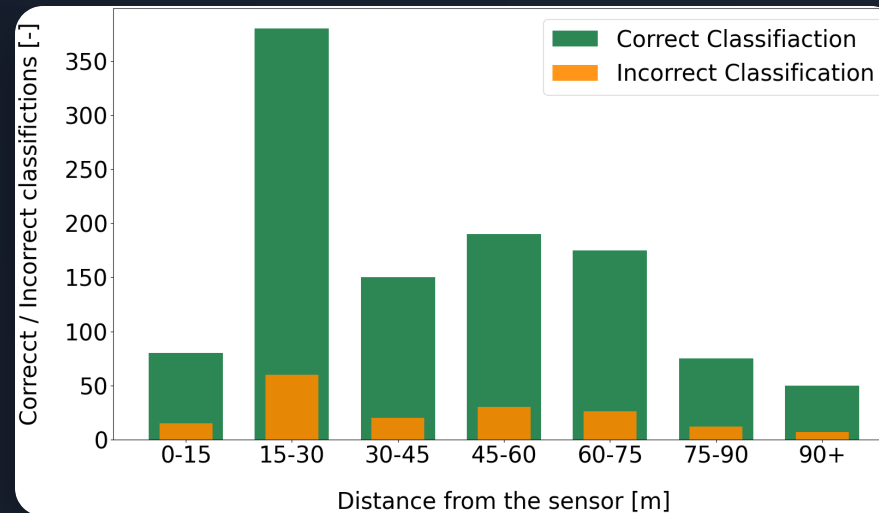


Fig. 4. The number of correct and incorrect classifications on the test dataset plotted against the object's distance from the sensor. As illustrated above, the distance only slightly affects the ratio of correct to incorrect classifications. This behaviour is the result of our collected samples being sparse to begin with, which greatly aided our classifier in generalizing for longer distances during training.

# Conclusions and Future Work

- Proposed method enables real-time UAV and bird detection using sparse LiDAR
- Attention-based segmentation and PointNet-based classification are effective
- Dynamic background removal outperforms static approaches
- Future: Analyze minimum reflection requirements and failure cases
- Plan to assess correlation between distance and reliability in-depth